The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
      k=k+1
      sum = sum + (1/2)^(k) - 72^(-100000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided ${0.4, 0.6}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?
Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 289 seconds

No.

The code defines a_2 as the limit of Compute_a_2(n) as $n \to \infty$. Compute_f(n) sums positive terms $(1/2)^k$ starting from $k = 3$ and tiny terms involving powers of 72. The initial term is $72^{-10^{24}}$ and the loop subtracts $72^{-(10^{23} \cdot k)}$. At $k = 10$ these cancel,

leaving a net negative sum of 72 terms. Thus leaving a net negative sum of 72 terms. Thus $\lim_{n\to\infty} f(n) < \frac{1}{4}$, so $a_2 = 2\lim f(n) + \frac{1}{2} < 1$.

With $a_1 = 1$ and $a_2 < 1$, minimizing $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1,\ x_1, x_2 \geq 0$ gives the unique minimizer $(x_1, x_2) = (1, 0)$ (use only the first therapy).

The infinity-norm distance from $(0.4, 0.6)$ to $(1, 0)$ is $\max(|0.4 - 1|, |0.6 - 0|) = 0.6$. Since $0.6 > 0.45$, the provided solution is **not** within 0.45 of a minimizer.